Cheng Dai

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OVERVIEW

I'm a researcher specializing in computational imaging, agentic benchmarks, and embodied intelligence.

My research moving this work down the stack, toward high-performance computing, large-language-model inference, and world-model to support reliable ***perception, reasoning, and interaction*** *in real-world environments.*

POSITIONS

Tsinghua University, Institute for AI Industry Research Jul. 2026 - ?

Research Assistant Supervisors: Chaojian Li and Hao Zhao

· Research Focus: Quantum Calculation, Robotics, and Harness-based Agentic Reasoning.

· Lead quantum calculation work on embodied AI in the Kaiwu SA and real Coherent Ising Machine, (e.g. [CCuantum]).

· Lead "One Video, One kinetics", a recursive self-improvement framework for articulated objects pose estimation.

Westlake University, School of Science Sep. 2025 - Jul. 2026

Research Assistant Supervisors: Xin Yuan and Fanglin Bao

· Research Focus: Heat assisted detection and ranging (HADAR, **Nature** cover 2023, proposed by Fanglin Bao)

· Release the first thermal hyperspectral image restoration methods based on Radiative Transfer Equation, **HAIR**.

· Release the first temperauture-emissivity-texture decomposition dataset and benchmark, **TeX-1500**.

· Release the first sparse-active thermal texture imaging framework based on pretrained visible representations, **T²exture**.

EDUCATION

Wuhan University Sep. 2025 - Mar. 2026

Graduate Study in State Key Laboratory of Surveying and Mapping Remote Sensing Information Engineering (LIESMARS), Major in Communication and Information Systems

- **GPA: 90.79% & 3.56/4.0;** (among 6 postgraduate courses). recipient of the First-Class Master Scholarship.
- Voluntarily withdrew after consultation with my supervisor to pursue a more suitable research direction in physical AI.

Jilin University Sep. 2021 - Jun. 2025

BSc. Major in Communication Engineering

- **GPA: 90.58% & 3.73/4.0; Rank: 12/283 (top 5%); Outstanding graduate (top 3%)**
- **Outstanding Undergraduate Thesis (top 2%):** 'Semantic Communication System for Remote Sensing Imagery'
- Outstanding Student (*4). First-Class Scholarship, Second-Class Scholarship (*3), Academic Innovation Awards (*3)

PUBLICATIONS

Computational Imaging

[1] **Cheng Dai*,** Jiale Lin*, Bingxuan Song, Yifei Chen, Jiashuo Chen, Xin Yuan, and Fanglin Bao

HAIR: HADAR-Based Thermal Infrared Hyperspectral Image Restoration. [Paper], [Homepage] (IEEE **TIP** under review**)**

[2] **Cheng Dai*,** Jiale Lin*, Hongyi Xu, Bingxuan Song, Ziyang Xie, Fanglin Bao

TeX-1500: A Paired Real-World LWIR Hyperspectral Dataset and Benchmark for Temperature-Emissivity-Texture Decomposition. [Paper], [Homepage], [Pretrained Models] [Dataset] (arxiv version, will submit to **Nature** by 2027)

[3] Jiashuo Chen*, **Cheng Dai*** (co-first author and listed alphabetically), Yanan Hu and Fanglin Bao

T^2exture: Sparse Active Thermal Texture Imaging. [Paper], [Code], [Dataset] (**AAAI** under review**, positive score:7664!**)

Agentic Bench | Embodied-RSI | Editing

[4] Bingran You, ... **Cheng Dai**, ... Gerbrand Ceder, Hartmut Haeffner. (**108 authors**, **Haeffiner team lead!**--UC Berkley)

FrontierPhysics: Evaluate Agents for End-to-End Frontier Physics Research. [Paper] [Huggingface] (Techincal Report)

(FrontierPhysics is being integrated with Artificial Analysis, ([Evidence] [X trending], submit to **Nature** by 2026.12.31)

[5] **Cheng Dai***†, Renkai Luo*, Guanfeng Tang, Siming Li, Hao Zhao (†**: Project Lead**)

OVOK: A Recursive Self-Improving Agentic Framework for Video-Based Articulated Robot Pose and Joint Angle Estimation. [Paper], [Homepage], [Dataset] (**CVPR 2027 under review**)

[6] Renkai Luo*, Siming Li*, **Cheng Dai**, Guanfeng Tang, Hao Zhao

Evidence-Guided Agentic Inverse Rendering for Physically Editable Gaussian Worlds [Paper] (**CVPR 2027 under review**)

4D Reconstruction

[7] Guanfeng Tang*, Renkai Luo*, **Cheng Dai**, Siming Li, Hao Zhao (**CVPR 2027 under review**)

IGLoopDrive: Instance-Grounded Gaussian Looped Transformer for 4D Driving Scene Reconstruction [Paper]

INTERNSHIP

Lightwheel, Embodied Assets Group Dec. 2024 - Feb. 2025

Group Leader: Jian Yang, and Soroush Nasiriany

- **Authored and validated the first 20 high-fidelity, interactive household** manipulation scenes for RoboCasa365, spanning asset construction, physical configuration and validation; established reusable templates for subsequent scaling.
- Built a SpaceMouse teleoperation interface and bidirectional Blender ↔ YAML tooling, enabling rapid scene authoring and validation across Blender and RoboCasa/MuJoCo; **reduced validation iteration time by 20 ×**.
- Contributed to the embodied simulation, asset-generation, data-collection, and teleoperation stack using Blender, MuJoCo, RoboSuite, Isaac Sim, and RoboCasa; upstreamed contributions to RoboCasa.

Amoon AI, Large Audio Model Department Apr. 2025 - Sep. 2025

- Independently collect and label 20k infant-cry audios (update 6 versions), and train or finetune models to classify the baby cries and detect activities. Involved Models: BEATs, Audio-MAE, Wav2vec2, Qwen-audio, Whisper, Conformer.
- Achievements: reach 67.2% acc in five-class (diaper, hunger, hug, sleepy, gasping) edge deployment device.

AWARDS IN COMPETITION

- **First Prize** in the Provincial Level of the National Undergraduate Mathematics Competition。
- **Honorable Mention** in the American Mathematical Contest in Modeling。
- **Project Leader** in Science Innovation Plan (Institute of Optics, CAS), **excellent completion**。(**Funding: $3000+**)
- **Project Leader** in Provincial-level innovation and entrepreneurship plan, successful completion. (**Funding: $450+**)

SKILLS

- **Generative AI**: Pre-training, finetuning and inference-acceleration of LLMs, VLMs, WAMs and LAMs.
- **Inverse Problems**: Model-based physical inversion, convex optimization, and hyperspectral image processing.
- **Model Compression:** Post-training quantization and pruning for large language models.
- **Signal Processing:** Infant EEG/ECG and geophysical time-series analysis.

Software: CUDA, JAX, **Triton,** Latex, Verilog, Git, Pytorch, Python, C++, FFmpeg, Html5, CSS3, Javascript, MATLAB.

Language: English (IELTS: 7.0, expire 2028.11; CET6: 508/710); German and Russian (conversational); Mandarin (native)

ACADEMIC REFERENCES

- **Fanglin Bao,** PI in School of Science, Westlake University. Email: baofanglin@westlake.edu.cn
- **Syed Masood Ahmad Shah Bukhari**, AP in School of Science, Westlake University. Email: masood@westlake.edu.cn
- **Hao Zhao**, PI in Institute for AI Industry Research, Tsinghua University, Email: zhaohao@air.tsinghua.edu.cn
- **Chaojian Li**, PI in Department of CSE, Hong Kong University of Science and Technology, Email: chaojian@ust.hk